
ADVANTEST®

ADVANTEST CORPORATION

ASX/U-51

Basic System Software

Software Specifications

MANUAL NUMBER 8256273-14

Applicable Products

*PASX51-93*****

*PASX51-92*****

*PASX51-91*****

*PASX51-85*****

*PASX51-81*****

*PASX51-77*****

*PASX51-75*****

*PASX51-71*****

*PASX51-70*****

*PASX51-36*****

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1. *Software Specifications*

This manual describes the ASX/U-51 system software.

1. 1 **Features of ASX/U-51**

ASX/U-51 is an exclusive operating system for T5500 series VLSI test systems. The ASX/U-51 system software runs on Solaris which is an operating system provided by Sun Microsystems Inc. and is compatible with the conventional MOST environment.

General operations as a computer such as file manipulation and network operation can be executed in Solaris environment and VLSI test system operation can be executed in MOST environment. ASX/U-51 thereby may be called a new generation operating system suitable for VLSI test systems because a suitable environment can be provided for each operating stage.

ASX/U-51 incorporates MOST environment as mentioned above, so that the conventional operation, know-how, and software resources of MOST can be utilized as they are.

ADVANSite employs two processors called the test processor and the environment processor, so that the test time is reduced and throughput and operability of program development in a background are improved. Users can use two processors as if they were using an environment processor.

To fully utilize the excellent ADVANSite environment, ASX/U-51 includes the software needed to operate two processors.

The T5500 series ATLworks is a new software tool used to evaluate and analyze memory devices efficiently with a graphical user interface. ATLworks allows even beginners to easily handle the increasingly complex test technology. And the ATLworks can effectively perform online tasks such as memory device evaluation. ATLworks uses Motif as a graphical user interface to provide a consistent interface experience.

Many optional software have been provided for the T5500 series ADVANSite test systems. They are the expanded functions from the MOST environment. Excepting some functions such as CLI, KED and so on, conventional usage is possible with the ASX/U-51 system software.

1. 2 Scope and Products Defined by this Specifications

This manual describes the ASX/U-51 system software.

The following software products are supported on T5500 series ADVANsite test systems.

◆ System Basic Software

T5500 series ADVANsite system software has the following types:

ASX/U-51	PASX51-93****	T5593
	PASX51-92****	T5592
	PASX51-91****	T5591 and T5591R
	PASX51-85****	T5586 and T5585
	PASX51-81****	T5581 and T5581H
	PASX51-71****	T5581P and T5571P
	PASX51-77****	T5377
	PASX51-75****	T5376 and T5375
	PASX51-70****	T5371
	PASX51-36****	T5336 and T5311

◆ T5500 series ADVANsite Optional Software

T5500 series ADVANsite optional software has the following types:

XATL/U-51:	PXTL51-00****	
	PXTL51-36****	
TESIM/U-51	PTSM51-81****	
MRA3:	PMR351-81****	
MRA4:	PMR451-71****	
MRAlite:	PMRL51-36****	
Workbench-M2:	PWBM51-00****	
Wave Tracer2:	PWTR51-00****	
MTBuilder:	PMTB51-81****	
MPATsim:	PMPS51-00****	
TMVERIFY/U-51:	PTTY51-00****	
WFBMAP3:	PWV00-00****	Wafer fail bit map display section
	PWFR51-00****	WFPHEd3 display data acquisition section
TCC/U-51:	PTCC51-00****	
SBOX/U-51:	PSBX51-00****	

➔ For more information on optional software, refer to each optional software specifications.

1. 3 Functions of ASX/U-51

◆ Inheritance of the MOST Environment

ASX/U-51 inherits the environment of MOST which is a conventional ADVANTEST standard operation system. ASX/U-51 allows you to create, compile, execute, and debug programs as before. Conventional utility programs can also be used on ASX/U-51.

ASX/U-51 has an upper-compatibility for programs created in ATL language.

◆ Multi-window Function

ASX/U-51 can display multiple emulation terminal windows on the large monitor screen and can be operated by switching those windows. It has the equivalent effect of using multiple terminals simultaneously and improves operationality.

◆ Function of Hierarchical File Management

ASX/U-51 provides file management function of hierarchical tree structure.

◆ Network Function

ASX/U-51 provides network function as a standard function. TCP/IP is used for network software. TCP/IP allows you to log in to remote node computers with the virtual terminal function, transfer files, and access to remote node files.

The MAIL function can also be used to interactively transfer text files.

◆ Language for Environment Processor Operation

ASX/U-51 uses C-shell as an interface to operate the environment processor.

C-shell is a command language interpreter having consistent syntax and expression.

C-shell has the command line editing function and the recall function.

◆ Text Editor

ASX/U-51 provides the vi editor as a standard text editor to create programs.

◆ System Security Function

ASX/U-51 has system security functions used to protect system programs, to adjust system resources, and to prevent against intrusion to user computers.

◆ Debugger

This function is effective to debug a program written in ATL language.

◆ System Diagnostic Program

Checks whether hardware of the test system operates normally.

◆ Help Function

The UNIX command online help (man) function is provided.

◆ ATLworks^{*1}

ATLworks has the following functions:

Toolbox

Activates each tool.

Control panel

Displays the control box settings on the screen.

Loads, activates, and stops device programs and sets data logs on the screen.

Debugger & editor

Edits, compiles, and debugs device programs (for example, to make breakpoint settings, reference to variables or variable assignments) on the screen.

Online manual

Displays an online manual in HTML or PDF format on the screen.

Test condition monitor

Displays hardware information on the screen.

Time scope tool

Measures the execution time of each device program test and displays this information on the screen.

DBM tool

Displays, debugs (displaying and modifying fails), and saves DBM pattern programs on the screen.

**1 ATLworks is bundled with ASX/U-51 system software REV 5.05 and later.*

◆ Hybrid Controller

The hybrid controller can execute device programs written in ATL, C, or C++ language.

◆ Online Manual

Online manual in HTML or PDF format is available on either of a graphic terminal screen or on a PC.

1. 4 Scope of Support

1. 4. 1 Test Systems Supported by ASX/U-51

ASX/U-51 can be use for the following VLSI test systems which are computable with the ADVAN-site upgrade.

T5581	ASX/U-51 REV 1.00 and later
T5581H	ASX/U-51 REV 1.00 and later
T5581P	ASX/U-51 REV 2.00 and later
T5571P	ASX/U-51 REV 2.00 and later
T5311	ASX/U-51 REV 3.00 and later
T5336	ASX/U-51 REV 3.00 and later
T5591	ASX/U-51 REV 4.00 and later
T5591R	ASX/U-51 REV 4.03 and later
T5371	ASX/U-51 REV 5.00 and later
T5585	ASX/U-51 REV 5.01 and later
T5592	ASX/U-51 REV 5.02 and later
T5586	ASX/U-51 REV 5.04 and later
T5375	ASX/U-51 REV 5.05 and later
T5376	ASX/U-51 REV 6.00 and later
T5593	ASX/U-51 REV 6.01 and later
T5377	ASX/U-51 REV 6.03 and later

1. 4. 2 Optional Hardware Supported by ASX/U-51

ASX/U-51 supports the following optional hardware:

Test System

Two-station configuration

Environment processor system

➔ Refer to the product specifications described in [1. 4. 1 "Test Systems Supported by ASX/U-51" on page 1-6.](#)

1. 5 Requirements

1. 5. 1 Hardware Requirements

The test system is either one of the systems shown in [1. 4. 1 "Test Systems Supported by ASX/U-51" on page 1-6](#).

Test systems which are compatible with Solaris 8.

T5593, T5592, T5586, T5585, T5377, T5376, T5375, and T5371 test systems

T5591, T5591R, T5581P, and T5571P test systems equipped with the V0713, V0713A, or V0714 (Sun Ultra 5)

Test systems which are compatible with the hybrid controller

T5593, T5592, T5586, T5585, T5377, T5376, T5375, and T5371 test systems



Limitation

Note that a 128 MB add-on memory board should be installed in the T5592, T5586, T5585, and T5371 test systems.

1. 5. 2 System Expansion

New devices and software may be needed in the future, depending on the software and/or hardware expansion.

1. 5. 3 System Requirement for ATLworks

◆ Hardware

[Table 1-1](#) shows the hardware required for ATLworks.

Table 1-1 Hardware Required for ATLworks

Item	Requirement			
Test system ^{*1}	➔ Refer to 1. 4. 1 "Test Systems Supported by ASX/U-51" on page 1-6 .			
EWS memory ^{*2}	For SPARCstation 5: 32 MB	For Sun Ultra 5: 64 MB	For Sun Blade 100: 256 MB	For Sun Blade 150: 256 MB
Hard disk ^{*3}	At least 20 MB of free disk space in the /export/home partition after installation.			

^{*1} The T5336 and T5311 are not included.

^{*2} ADVANsite satisfies this requirement. However, ADVANTEST recommends you to allocate more memory space than the requirement size because ATLworks uses various graphic applications.

^{*3} Approximately two times as much free space is required when installing this software.

◆ Software

Table 1-2 to Table 1-4 show related software required for UNIX operating systems.

Table 1-2 Solaris 2.5.1 (November 1997 or Later Versions)

Item	Requirement
X11 server	OpenWindows V3.5.1
Window manager	Motif V1.2.3

Table 1-3 Solaris 7

Item	Requirement
X11 server	CDE V1.3
Window manager	Workspace Manager

Table 1-4 Solaris 8

Item	Requirement
X11 server	CDE V1.4
Window manager	Workspace Manager

1. 5. 4 System Requirement for Online Manual

Hard disc requirements for using manuals in HTML format

Solaris edition

Netscape Navigator 6.2.3	47 MB
Acrobat Reader 4.05	17 MB
Namazu 2.0.12	4 MB

Windows edition

Netscape Navigator 6.2.3	20 MB
Acrobat Reader 5.1 for Windows	26 MB

**1 Online manuals in HTML format consist of both HTML and PDF formats.*

Hard disc requirements for using manuals in PDF format

Solaris edition

Acrobat Reader 4.05	17 MB
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Windows edition

Acrobat Reader 5.1 for Windows	26 MB
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1. 6 Software Support

Software support is free of charge for one year after delivery.

Terms and conditions of the software support is stipulated in the software support standard of ADVANTEST corporation.

Software support after one year is available at cost.

It is recommended to conclude software support contract after one year warranty elapsed in order to keep the software up to date.

However, the software support by ADVANTEST does not cover support for the Tornado software bundled with ASX/U-51 system software. To receive Tornado software support, conclude software support contract with Wind River Systems, Inc., or the authorized sales agency.

1. 7 Order

When ordering software, inform us of the present environment, number and type of the connected test system besides the name of the software.

Especially, the above mentioned information is necessary to install ADVANsite.

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